

1)

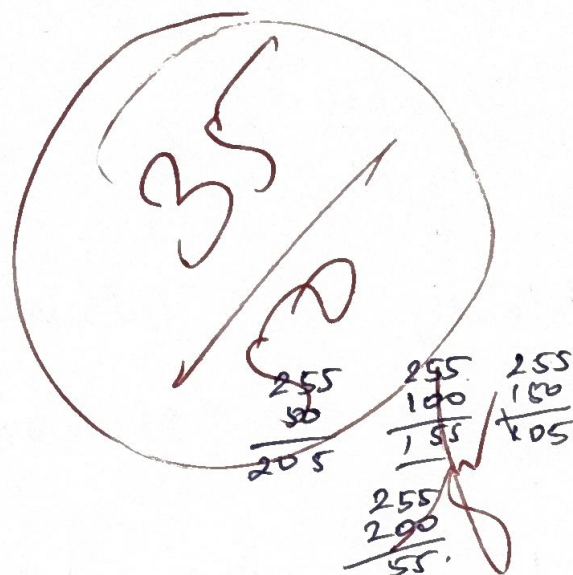
(a) Negative transformation

$$s = 255 - r$$

Original Pixel

Transformed Pixel

50	205
100	155
150	105
200	55



(a) Grey level transformation in image Enhancement

It is used to improve the image appearance by changing Pixel intensity values.

(2) Histogram Equalization.

Intensity	frequency.
0	4
1	6
2	10
3	6

(b) CDF = ?

Intensity	frequency	CDF
0	4	4
1	6	10
2	10	20
3	6	26

PDF?
formula?

(1) Equalized intensity Value

$$S = \frac{CDF}{255} \times 255$$

Intensity Calculation

$$0 \quad \frac{1}{255} \times 255 = 0.46$$

$$1 \quad \frac{10}{255} \times 255 = 1.5$$

$$2 \quad \frac{20}{255} \times 255 = 2.30$$

$$3 \quad \frac{25}{255} \times 255 = 3$$



(2) Histogram Equalization

It improves image contrast by spreading pixel intensity value over the full intensity range.



3)

(a) Gaussian filtering is preferred over averaging filtering.

It preserves better edge when compared to average filter.

It gives the weight to the centre

It reduces the noises.

(b) Apply Gaussian Kernel

$$\frac{1}{16} \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix}$$

$$= \frac{1}{16} \times$$

Image region

$$\begin{bmatrix} 10 & 20 & 10 \\ 20 & 40 & 20 \\ 10 & 20 & 10 \end{bmatrix}$$

$$= \frac{1}{16} (10 + 40 + 10 + 40 + 10 + 40 + 10 + 40 + 10)$$

$$= \frac{1}{16} (360)$$

$$= \frac{360}{16} = 22.5$$

$$= 22.5 \approx 23$$

$$\begin{array}{r} 22.5 \\ 16 \overline{) 360} \\ \underline{32} \\ 40 \\ \underline{32} \\ 80 \\ \underline{80} \\ 0 \end{array}$$

$$\begin{array}{r} 16 \\ 16 \\ \underline{32} \\ 3 \\ 16 \times 5 \\ \underline{80} \end{array}$$

(4)

(a) Concept of edge detection.

Edge detection is nothing but it identifies object boundaries by detecting sudden intensity changes.

(b) Horizontal Sobel mask,

$$\begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix}$$

Image.

$$\begin{bmatrix} 10 & 10 & 10 \\ 20 & 20 & 20 \\ 30 & 30 & 30 \end{bmatrix}$$

$$\text{Gradient} = (-1 \times 10) + (-2 \times 10) + (-1 \times 10) + (1 \times 30) + (2 \times 30) + (1 \times 30)$$

$$= -10 - 20 - 10 + 30 + 60 + 30 = -40 + 120 = 80$$

Gradient = 86.

5) (b) Linear contrast stretching to map the range [50, 100] to [0, 255]
r1 = 50, 100, 150.

$$s = \frac{r - r_1}{r_2 - r_1} \times 255$$

$$s = \frac{r - 50}{100} \times 255$$

When $r = 50$

$$s = \frac{50 - 50}{100} \times 255$$

$$s = 0$$

When $r = 100$

$$s = \frac{100 - 50}{100} \times 255$$

$$s = \frac{50}{200} \times 255$$

$$s = \frac{225}{2}$$

$$s = 112.5$$

$$s = 113$$

Original Value

New value

50

0

100

113

150

255

$$\begin{array}{r} 112.5 \\ 2 \overline{) 225} \\ \underline{22} \\ 05 \\ \underline{04} \\ 10 \\ \underline{10} \\ 0 \end{array}$$

When $r = 150$

$$s = \frac{150 - 50}{100} \times 255$$

$$= \frac{100}{100} \times 255$$

$$s = 255$$

(a) Contrast stretching

This will increase the D/B dark and bright pixels and making the image more details and visible